Table 3.2: Difference between Bactericidal and Bacteriostatic

Bactericidal	Bacteriostatic
Bactericidal refers to antibiotics that kill bacteria	Bacteriostatic refers to antibiotics that prevent the growth of bacteria
Action is irreversible	Action is reversible
Inhibit the cell wall formation of bacteria	Inhibit DNA replication and protein synthesis of bacteria
Do not work with the immune system of the host	Work with the immune system of the host to prevent the growth and reproduction of bacteria
MBC refers to the concentration of the drug required to kill 99.99% of the bacterial population	MIC is the minimum drug concentration which inhibits the bacterial growth
Examples include betalactam antibiotics, cephalosporins, and vancomycin	Examples include tetracyclines, spectinomycin, chloramphenicol, sulfonamides, etc.

3.2 Factors Influencing the Antimicrobial Activity of Chemical Agents

The following factors will affects the activity of a disinfectant or antiseptic and these should be borne in mind during use.

a. The Concentration and kind of a chemical agent used

The higher the concentration of the germicide the greater will be the rate of killing. This is particularly important with the phenolic group of compounds, whose activity falls off very rapidly with dilution.

b. Time of exposure to the agent

In general germicidal activity is increased with time and a sufficient exposure is imperative for efficient disinfection.

c. Temperature at which the agent is used

An increase of temperature will also raise the rate of killing.

d. Presence of Organic matter

Most germicides are reduced in activity by the presence of organic matter and particularly by the presence of proteins such as those in body fluids.

e. Number of organisms present

The larger the number of organisms, the greater will be the time required for disinfection.

f. The kinds of microorganisms present - Presence of spores

Spores are exceptionally resistant to the great majority of disinfection.

g. Mode of action of antimicrobial agent

The nature of the material bearing the microorganism.

3.3 Mode of Action of Chemical Agents

Chemical agents act on microorganisms by:

- They may damage the lipids and proteins of the cytoplasmic membrane of microorganisms.
- They may denature microbial enzymes and other proteins usually by disrupting the hydrogen and disulfide bonds that give the protein its 3-D shape. This blocks metabolism function.

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